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PROFESSIONAL SUMMARY

M.Sc. Cybersecurity student and Research Assistant at CISPA and Saarland University ML Team, with research experience spanning **AI Security**, **Web Security**, **Explainable AI**, and **Trustworthy Machine Learning**. Currently leading research on Service Worker-mediated Web Cache Deception (SW-WCD), autonomous coding agent security, and concept-based interpretability methods for deep learning systems. Experienced in designing large-scale experimental infrastructures, evaluating AI system reliability and vulnerabilities, developing reproducible ML pipelines, and conducting empirical security and machine learning research.

TECHNICAL SKILLS

Web & Offensive Security: VAPT, System exploitation, Privilege escalation, Proof of concept development, Client-side vulnerability assessment, Browser and extension testing, CDN Architecture (Nginx/Varnish/Cloudflare), OWASP Top 10.

Security Operations: Splunk, Wireshark, Autopsy, MITRE ATT&CK, ITIL Framework, ISO 27001, Security Audits, Risk Assessment.

Cloud & DevSecOps: AWS Security fundamentals, Azure SIEM/SOAR, Infrastructure as Code (Terraform), Container security (Docker, Kubernetes), CI/CD Security concepts, Misconfiguration analysis.

Programming: Python (automation, security tooling, agent workflows), Bash/Shell Scripting, JavaScript/Node.js, C++

ML & Tools: PyTorch, TensorFlow, Autoencoders, RL, LLMs (Claude Code), HuggingFace, RunPod, Langchain, VectorDatabases, Docker, Kubernetes.

AI Security: Attacks : (Prompt Injection, Jailbreaking, Data Poisoning, Membership Inference attack. Defense : (Input Sanitization (Guardrails), Adversarial Training, Robustness Verification.)

AI & Security Automation: AI-assisted security workflows, Integration with LLMs (Claude, GPT, Llama), agent-based automation.

Database & Data Handling: SQLite3, PostgreSQL, JSON-based data modeling, dataset preprocessing for security and ML pipelines.

Productivity Ecosystem: Workspace (Script automation, Data Studio reporting), Jira (Agile workflow management), Documentation (GitHub, Markdown), Communication (Slack, Teams, LaTeX).

PROFESSIONAL EXPERIENCE

Saarland University – Machine Learning Group

Research Assistant (Explainable AI & Trustworthy Machine Learning)

Saarbrücken, Germany

April 2026 – Present

- **Concept Bottleneck Models (CBMs):** Conducting research on spatially intervenable Concept Bottleneck Models using the CUB-200 dataset, investigating how concept-level interventions influence downstream model predictions and interpretability.
- **Concept Segmentation :** Built an automated concept-grounding framework leveraging Segment Anything Models (SAM) by META to generate concept-specific spatial masks and evaluate concept localization quality across fine-grained visual recognition tasks.
- **Intervention Faithfulness Analysis:** Designed and executed large-scale Mask-IN and Mask-OUT intervention experiments to quantify the causal impact of concept regions on concept activations and classification performance, evaluating concept faithfulness beyond traditional accuracy metrics.
- **Saliency-Guided Interventions:** Investigating whether continuous saliency representations can serve as effective intervention mechanisms, comparing saliency-based interventions against explicit concept interventions to study representation faithfulness and model behavior.

CISPA – Helmholtz Center for Information Security

Research Assistant (Web & AI Security)

Saarbrücken, Germany

Sep 2025 – Present

- **SW-WCD Research:** Developed SW-WCD-RESEARCH, a controlled prototype to study Service Worker-mediated Web Cache Deception. Architected a simulation environment using Nginx (CDN logic) and Express (Origin) to test path sculpting and header manipulation payloads.
- **Infrastructure Automation:** Implemented a Playwright-based test suite for cross-browser validation and an R-based statistical engine to compute attack success rates and time-to-cache metrics.
- **AI Agent Security (IDAD):** Leading the Vibecoding Security Gap study, benchmarking Agent-Native IDEs vs. CLI Agents. Designing the XYZ Bench framework to evaluate 275 development tasks for hallucinated dependencies, logic optimization bypasses, and security vulnerabilities.
- **Vulnerability Discovery:** Uncovering novel Verification Gap, risks where autonomous agents report successful tests on vulnerable code, specifically in context poisoning and legacy pattern propagation & identifying Security Vuln.

Hackers4u

Penetration Testing Intern

Remote

Sep 2023 – Oct 2023

- Executed black-box penetration tests on web applications, identifying critical vulnerabilities (SQLi, XSS, IDOR).
- Automated network enumeration using Nmap and Nessus APIs, reducing manual scanning time by 40%.
- Documented findings in markdown reports for technical and non-technical stakeholders.

CybersecuredIndia

Cybersecurity & Digital Forensics Intern

Bangalore, India

Dec 2022 – Feb 2023

- Analyzed malware behavior logs using Python to detect obfuscation patterns.
- Conducted forensic investigations preserving chain-of-custody; reconstructed attack timelines using Registry and Prefetch artifacts.
- Translated findings into actionable risk mitigation plans.

Virtual Testing Foundation

Information Security Administrator Intern

Remote
Sep 2022 – Nov 2022

- Mapped adversary behaviors to controls using MITRE ATT&CK to identify gaps.
- Assisted in drafting Information Security and Acceptable Use Policies (ISP/AUP).

KEY PROJECTS & RESEARCH

SW-WCD-RESEARCH: Web Cache Deception Prototype | *Docker, Nginx, Playwright, R, JS, SQL* Research

- Engineered a complete research testbed to evaluate how Service Workers influence WCD behaviors in CDN-backed architectures.
- Implemented Node.js anomaly detectors to log rewritten URLs and cache indicators.
- Developed attack payloads (e.g., t1-path-sculpting.js) to bypass standard CDN cache armor.
- Designed PostgreSQL schema for trial data storage and statistical power analysis.

The Vibecoding Security Gap (IDAD Evaluation) | *Python, eBPF, Gemini 3.5, Claude Code* Research

- Building XYZ Bench to compare Agent-Native IDEs vs. CLI agents across 275 software tasks.
- Measuring hallucinated dependencies and context poisoning rates in autonomous coding.
- Using eBPF to monitor insecure execution patterns in agent-generated code.

Protocol Fuzzing | *Python, Fandango, SMTP, Redis-like Protocol* Course Project

- Designed a stateful Fandango IO grammar for a Redis-like key-value store, enabling valid command sequences (SET/GET/UPDATE) with response validation.
- Simulated an SMTP man-in-the-middle attack by hijacking authenticated sessions to send spoofed emails without hardcoded credentials.
- Used derivation tree constraints and prefix-aware tracking to enforce protocol correctness during fuzzing.

HACKBOT: AI-Powered Automated Exploit Engine | *Python, Meta-Llama 3, RunPod, RAG* GitHub

- Local LLM interface with RAG to query CVE databases without hallucinations.
- Deployed on RunPod with latency/resource optimization.
- Integrated static analysis for auto-generated vulnerability reports.

API-Based NMAP Dashboard | *Python, Flask, REST API* GitHub

- Full-stack dashboard for managing and visualizing Nmap scans.
- Enabled real-time asset visibility and centralized reporting.

PUBLICATIONS

Autoencoder-Driven Machine Learning for Advance Cybersecurity Malware Detection

FMDB Transactions on Sustainable Intelligent Networks, Vol. 1, No. 4, pp. 252–264, 2024.

DOI: 10.69888/FTSIN.2024.000292

Developed an autoencoder-based AI model to detect and classify malware from the EMBER dataset.

Docker Based Decentralized Vulnerability Assessment with Port Scanning

FMDB Transactions on Sustainable Intelligent Networks, 2024.

DOI: 10.69888/FTSIN.2024.000290

Integrated GPT-3 and Docker for efficient, distributed vulnerability scanning in decentralized systems.

API-based Network Scanning, Mapana

In: *Proceedings of National Conference on Emerging Trends in Computer Science*, 2024.

ISBN: 978-93-95830-43-0

Built a Python-Flask API for dynamic Nmap scan profiling and centralized dashboard monitoring.

Enhancing Fog Computing Through Data Center Expansion

SIJSS-UGC Journal (ISSN: 0972-8945), 2023.

Proposed a load-balancing algorithm improving IoT scalability by 40% via distributed resource optimization.

CERTIFICATIONS

Certified Ethical Hacker (CEH v12) - EC-Council

Certified Network Defender (CND v2) - EC-Council

IBM: Python for Data Science - IBM

IBM: Cybersecurity Analyst - IBM

Practical Penetration Testing - TCM Security